

Second part: Open book

Total score: 10 pts.

You are warmly invited to be tidy and concise. Messy papers will be penalized.

5. A transportation analyst has to process the data coming from the smart cards of the passengers. If everything works properly, a trip registration is a pair made of one input “stamp” (input station and time) and one output “stamp” (output station and time). However, sometimes this pairs are incomplete, thus either the input stamp or the output stamp is missing.

Let $I = \{1, \dots, n\}$ be the set of input stamps not associated with any output stamp. And let $O = \{1, \dots, n\}$ be the set of output stamps not associated with any input stamp. Associated with each $i \in I$ we have the time a_i and the station s_i . Associated with each $j \in J$ we have the time b_j and the station s_j . The analyst has to match the entering registrations in I with the exiting ones in O so as to complete the input-output pair in the most likely way. Knowing the average time c_{ij} to travel between station s_i and station s_j , the analyst has to match each $i \in I$ to a $j \in O$ so as to minimize

$$\max |c_{ij} - (b_j - a_i)|$$

Describe a linear model, specifying sets, parameter, variables (and their meaning) and constraints. Pay attention to the non linearities also in the objective function (the absolute value and the max are non linear).

6. A class of $n = 3k$ students has to be partitioned into k terns. For each pair of students i and j the compatibility level $c_{ij} \geq 0$ is known. The compatibility of a tern (i, j, h) is given by $c_{ij} + c_{ih} + c_{jh}$. The teacher has to find the partition of the class that maximizes the sum of the compatibilities over all terns. Describe a greedy algorithm. Specify the sets E, F and the functions $Best(S)$ and $Ind(X, e)$.